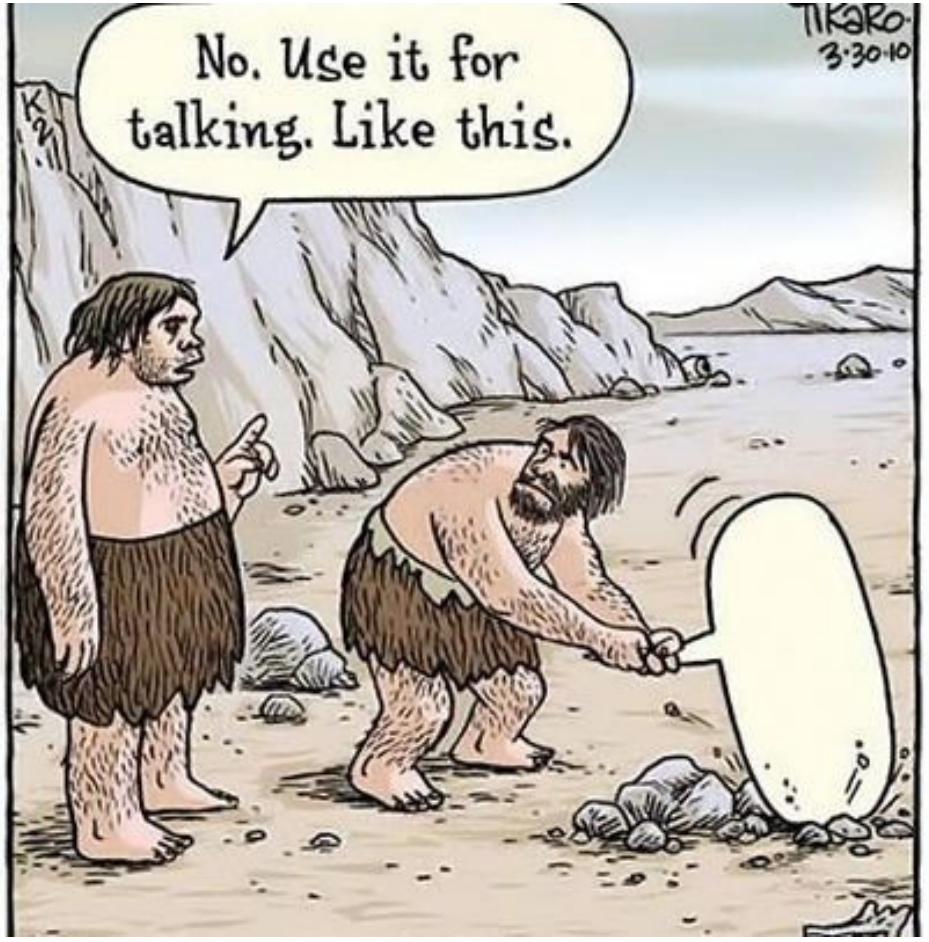
Natural Language Processing

Introduction, Terminology



Natural Language Processing

No unique reference book for the course: the slides + the pointers to material available on the web or on the AulaWeb module are enough for understanding the subject, even for students who cannot attend the frontal lessons.

Some lessons are based on the book: Speech and Language Processing (3rd ed. draft) by Dan Jurafsky and James H. Martin

https://web.stanford.edu/~jurafsky

Some slides integrate slides associated with that book.

Natural Language Processing

Natural language processing (NLP) is a subfield of linguistics, computer science, and artificial intelligence concerned with the interactions between computers and human language, in particular how to program computers to process and analyze large amounts of natural language data, to understand text (Natural Language Understanding, NLU), and to produce understandable texts in human languages from some underlying non-linguistic representation of information (Natural Language Generation, NLG).

Natural Language Processing

Symbolic NLP (1950s – early 1990s)

- Eliza (1966), https://en.wikipedia.org/wiki/ELIZA
- SHRDLU (1968-1970), https://en.wikipedia.org/wiki/SHRDLU
- PARRY (1972), https://en.wikipedia.org/wiki/PARRY
- Ontologies (MARGIE, 1973), https://www.ijcai.org/Proceedings/73

Natural Language Processing

Statistical NLP (1990s–2010s)

- Availability of large corpora, also multilingual
- Statistical models make soft, probabilistic decisions based on attaching real-valued weights to each input feature
- Example: Part-of-speech tagging based on Hidden Markov Models

Natural Language Processing

Neural NLP

- End-to-end learning of a higher-level task (e.g., question answering) instead of relying on a pipeline of separate intermediate tasks
- No need to engineer features that charaterize the language model
- New paradigm distinct from statistical natural language processing (example: neural machine translation)
- Example: word embedding

Natural Language Processing

Transfomers (it seems a long time ago...)

- BERT, Google (2019), Bidirectional Encoder Representations from Transformers,

  https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-

Natural Language Processing

Generative AI, LLMs (present)

- GPT, Generative Pre-Trained Transformer (ChatGPT released in November 2022)
- Llama, ollama, Gemini, ...

Natural Language Processing

Traditional NLP methods and approaches are required:

- when the amount of training data is insufficient to successfully apply machine learning/deep learning methods, or to train LLMs;
- when preprocessing in NLP pipelines, e.g., tokenization, or postprocessing and transforming the output of NLP pipelines, e.g., for knowledge extraction from syntactic parses, is needed; this includes refining/validating the results of ML/DL/LL models
- when LLMs cannot be used for technical, legal, ethical reasons

Natural Language Processing

We will discuss (some of) the common problems in the NLP area, following a standard linguistic analysis approach.

We will provide examples of (some of) the common techniques, and their common implementations.

Real example from our most recent research actvity, in the intersection of S&DAI and NLP

ChatBDI: The idea is very simple...

ChatBDI: The idea is very simple... think BDI, talk LLM!

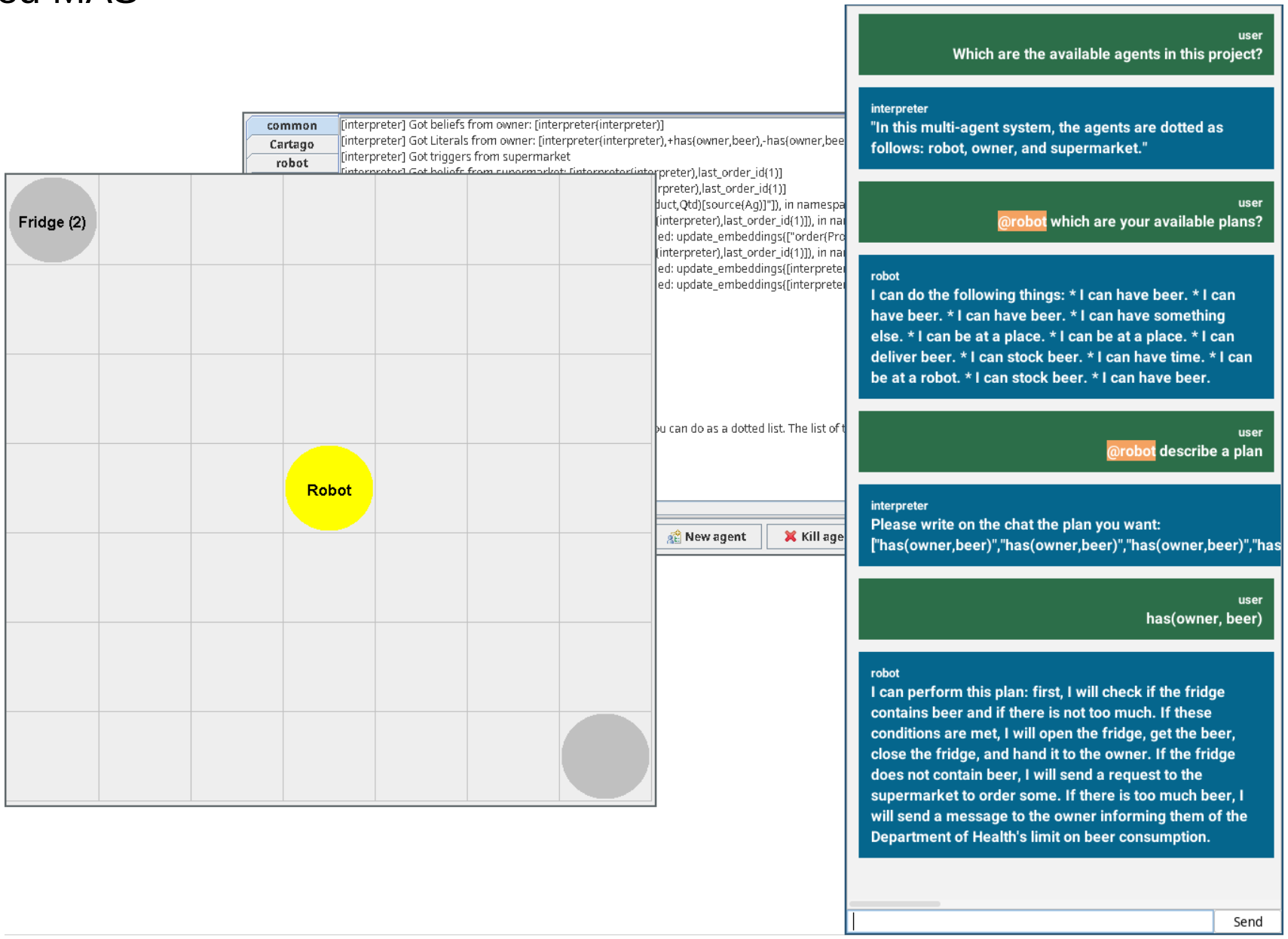
BDI brain



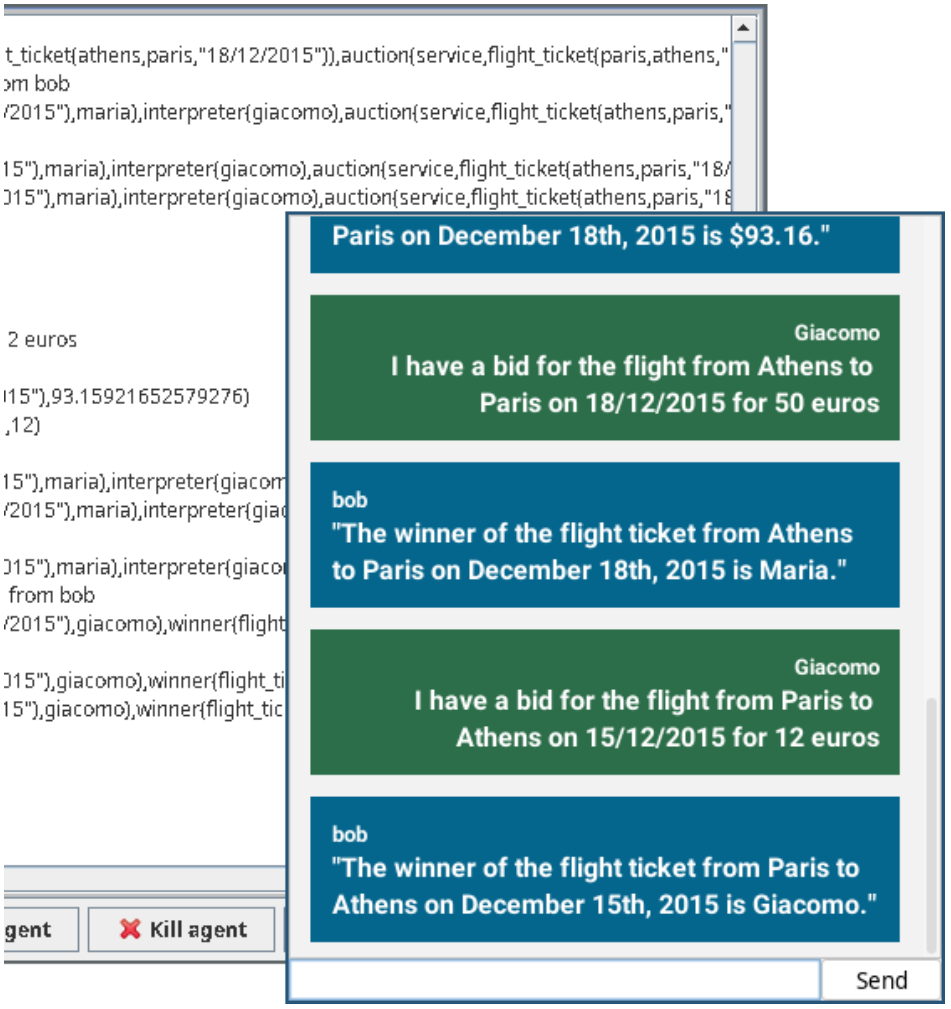


LLM language sensor-actuator

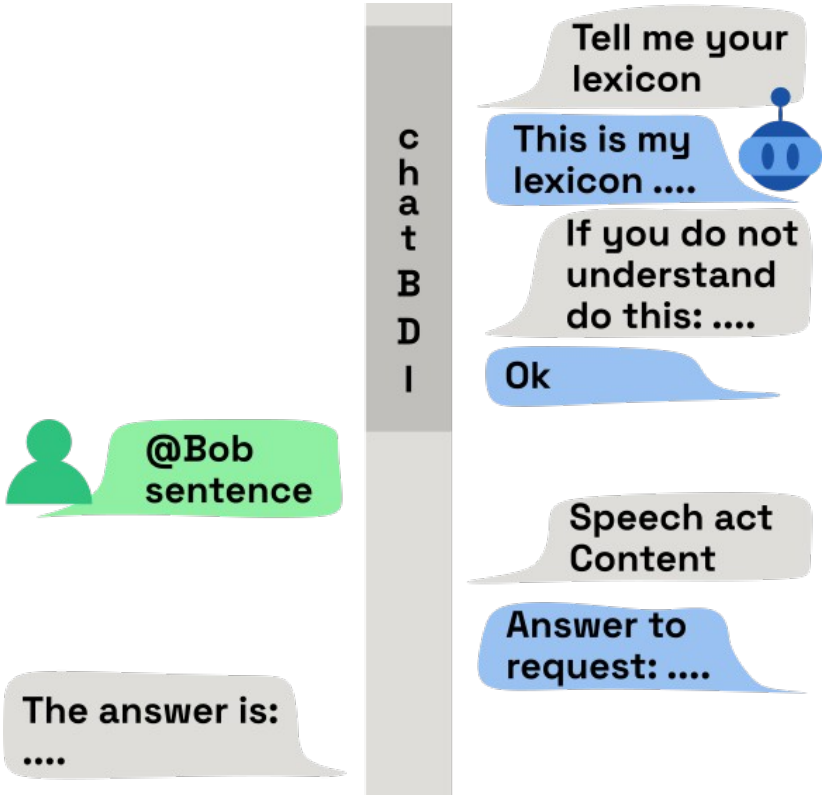
Scenario:
Closed MAS



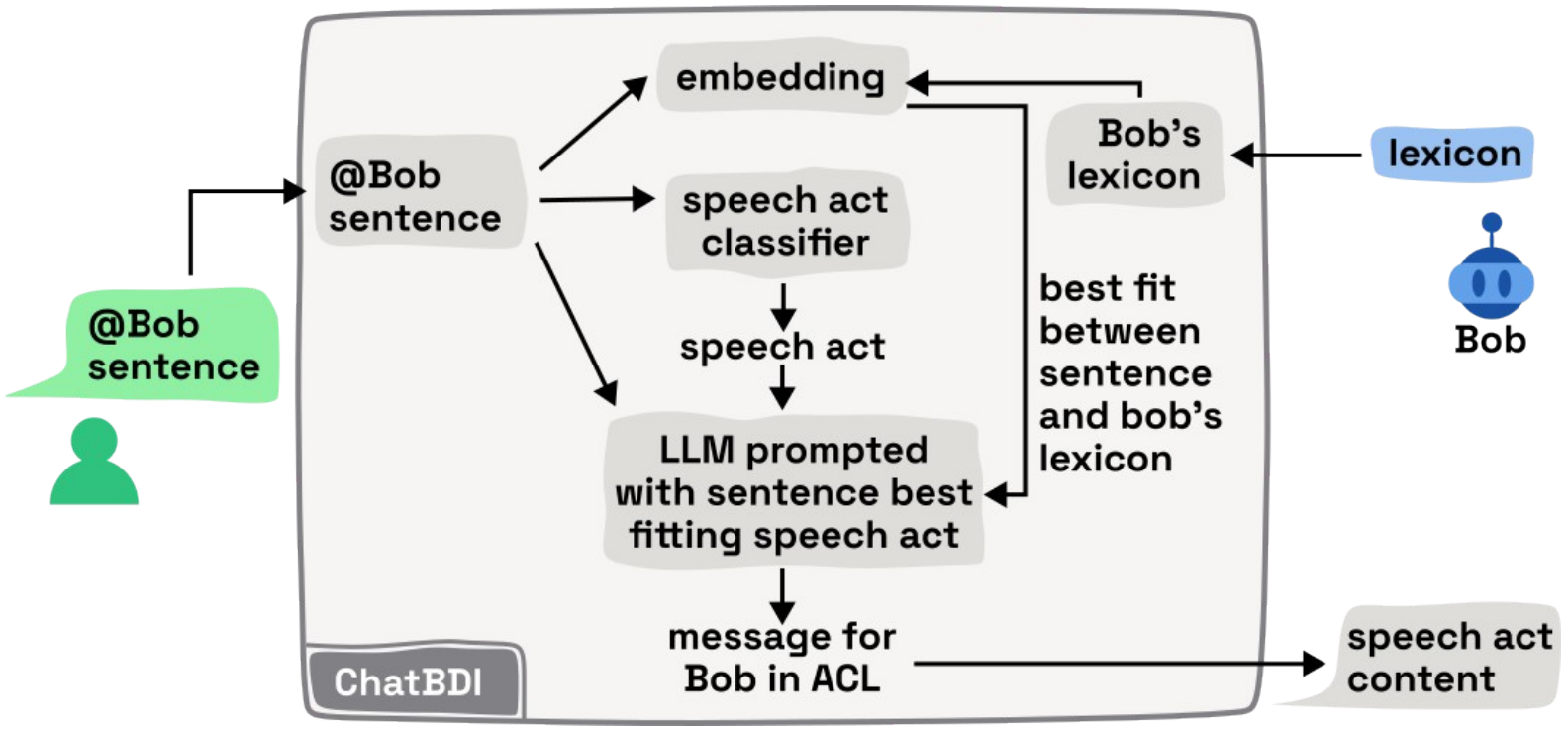
Scenario:
Open MAS



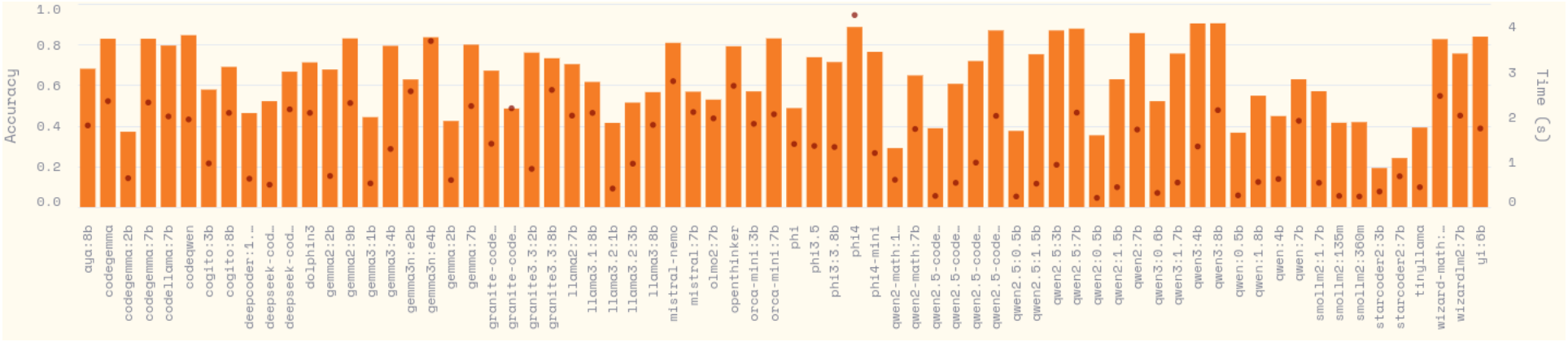
From an architectural point of view (kept simple)



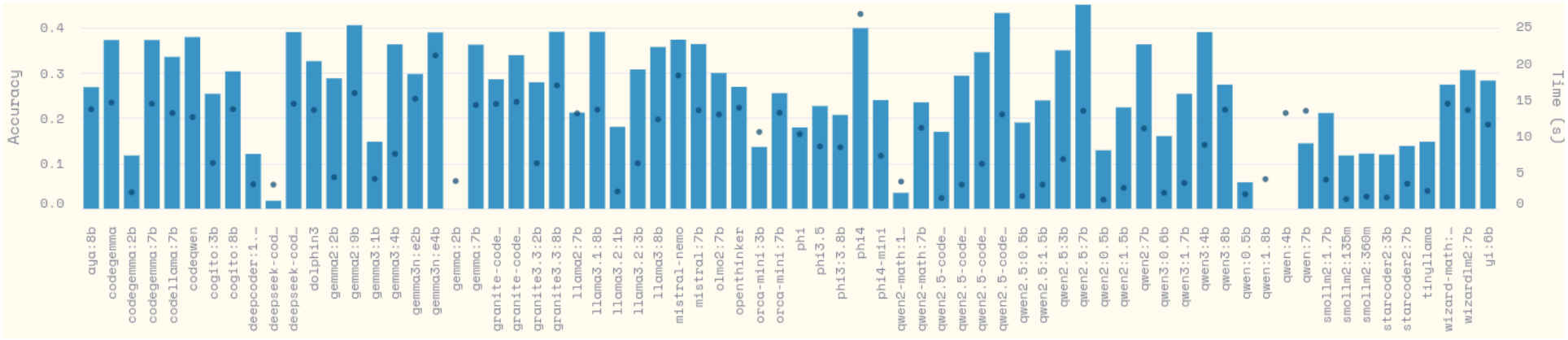
From an architectural point of view (kept simple)



Experiments: performatives



Experiments: message content



Well… it’s a hard problem to face...

Sentence

Partial

Employer 7 attend JS course till 1/11/2025

Performative	Expected
tell	tell

Term	Expected	Levenshtein
training_attended(7, JS, 1/11/2025)	training_attended(7, js, "1/11/2025")	4

Performative Time	Generation Time	Total Time
1.856 s	14.713 s	16.569 s

Well… it’s a hard problem to face...

Sentence

Partial

beans are very good for you, they contain 24 grams of protein per 100 grams

Performative	Expected
tell	tell

Term	Expected	Levenshtein
calories(beans, unit(24, gr), 100)	protein(beans, unit(100, gr), 24)	13

Performative Time	Generation Time	Total Time
2.316 s	19.339 s	21.655 s

Well… it’s a hard problem to face...

Sentence

Incorrect

Tell me the pressure of both front tires

Performative	Expected
tell	askAll

Term	Expected	Levenshtein
tire_pressure(_, 28)	tire_pressure(front, _)	7

Performative Time	Generation Time	Total Time
1.745 s	13.306 s	15.051 s

The three levels of linguistic analysis

(when in the exam I ask for the three levels of linguistic analysis, I refer to these ones...)

Levels of linguistic analyses

Pragmatics: what does it do?

Semantics: what does it mean?

Syntax: what is grammatical?

natural language utterance

Analogy with programming languages

Syntax: no compiler errors

Semantics: no implementation bugs

Pragmatics: implemented the right algorithm

Analogy with programming languages

Syntax: no compiler errors

Semantics: no implementation bugs

Pragmatics: implemented the right algorithm

Different syntax, same semantics (5):

$$2 + 3 \quad \Leftrightarrow \quad 3 + 2$$

Same syntax, different semantics (1 and 1.5):

$$3 \ / \ 2 \text{ (Python 2.7)} \quad \not\Leftrightarrow \quad 3 \ / \ 2 \text{ (Python 3)}$$

Good semantics, bad pragmatics:

correct implementation of deep neural network
for estimating coin flip prob.

NLP: Syntax

Regular Expressions

NLP: Syntax

Word Level

- Word segmentation
- Identification of stop words
- Stemming and Lemmatization
- Minimum edit distance

NLP: Syntax

Sentence Level

- Sentence breaking
- Part of Speech Tagging
- Parsing

NLP: Semantics

- Distributional semantics
- Lexical semantics
- Ontologies
- Word sense disambiguation
- Named entity recognition (NER)

NLP: Transformers and LLM

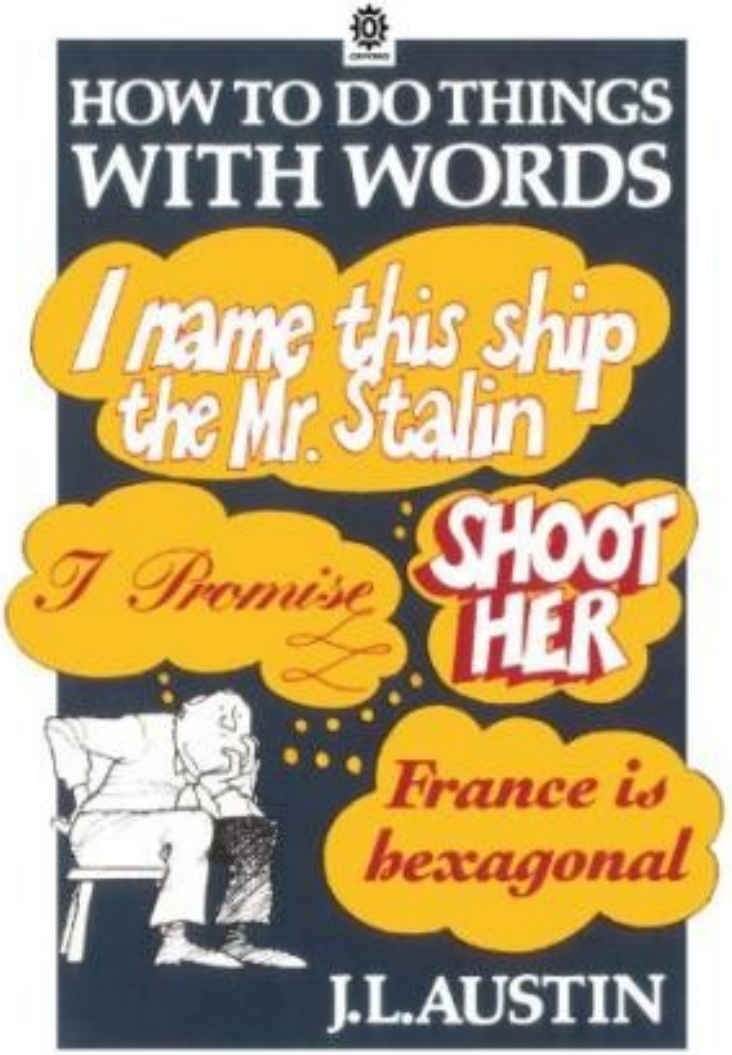
- BERT
- GPT

NLP: Most common (non trivial) features for representing a text

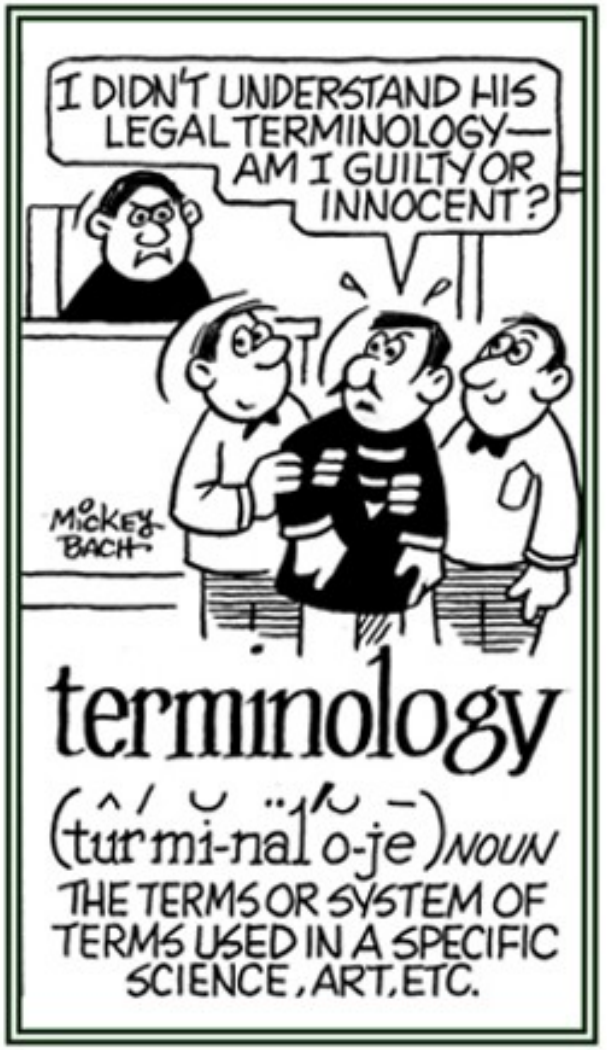
- Bag of Words
- N-grams (at word and char level)
- TF-IDF
- Flesch-Kincaid readability measure
- Linguistic inquiry and word count (LIWC)
- Emotion words (Ekman, Plutchik)
- .....

NLP: Pragmatics

- “How to do things with words”



Terminology



Terminology



Morpheme: A morpheme is the smallest grammatical unit in a language. In other words, it is the smallest meaningful unit of a language. The linguistics field of study dedicated to morphemes is called morphology. A morpheme is not identical to a word, and the principal difference between the two is that a morpheme may or may not stand alone, whereas a word, by definition, is freestanding.

Terminology



Every morpheme can be classified as either free or bound.

Free morphemes can function independently as words (e.g. town, dog).

Bound morphemes appear only as parts of words, always in conjunction with a root and sometimes with other bound morphemes. For example, un- appears only accompanied by other morphemes to form a word. Most bound morphemes in English are affixes, particularly prefixes and suffixes.

Example: "Unbreakable" comprises three morphemes: un- (a bound morpheme signifying "not"), -break- (the root, a free morpheme), and -able (a free morpheme signifying "can be done").

Terminology



Bound morphemes can be further classified as derivational or inflectional.

Derivational morphemes, when combined with a root, change either the semantic meaning or part of speech of the affected word. For example, in the word happiness, the addition of the bound morpheme -ness to the root happy changes the word from an adjective (happy) to a noun (happiness). In the word unkind, un- functions as a derivational morpheme, for it inverts the meaning of the word formed by the root kind.

Inflectional morphemes modify a verb's tense, aspect, mood, person, or number, or a noun's, pronoun's or adjective's number, gender or case, without affecting the word's meaning or class (part of speech). Examples of applying inflectional morphemes to words are adding -s to the root dog to form dogs and adding -ed to wait to form waited. An inflectional morpheme changes the form of a word.

Terminology



Word: a word is the smallest element that can be uttered in isolation with objective or practical meaning.

This contrasts deeply with a morpheme, which is the smallest unit of meaning but will not necessarily stand on its own. A word may consist of a single morpheme (for example: oh!, rock, red, quick, run, expect), or several (rocks, redness, quickly, running, unexpected), whereas a morpheme may not be able to stand on its own as a word (in the words just mentioned, these are -s, -ness, -ly, -ing, un-, -ed).

Terminology



Corpus (pl. Corpora): a text corpus is a large and structured set of texts (nowadays usually electronically stored and processed). They are used to do statistical analysis and hypothesis testing, checking occurrences or validating linguistic rules within a specific language territory.

Example: https://corpus.byu.edu/

Terminology

- **Genre:** a style or category of art, music, or literature.
- **Genre of a corpus:** usually defined on the basis of text-external features, such as medium, function and format.

Terminology



Collocation: A collocation is a sequence of words or terms that co-occur more often than would be expected by chance.

Some examples are "pay attention" ,"fast food", "make an effort", and "powerful engine".

Example: https://corpus.byu.edu/now/

natural English	**unnatural English**
the fast train	the ~~quick~~ train
fast food	~~quick~~ food
a quick shower	a ~~fast~~ shower
a quick meal	a ~~fast~~ meal

Terminology



Modifier	Head	Compound
noun	noun	football
adjective	noun	blackboard
verb	noun	breakwater
preposition	noun	underworld
noun	adjective	snow white
adjective	adjective	blue-green
verb	adjective	tumbledown
preposition	adjective	over-ripe
noun	verb	browbeat
adjective	verb	highlight
verb	verb	freeze-dry
preposition	verb	undercut
noun	preposition	love-in
adverb	preposition	forthwith
verb	preposition	takeout
preposition	preposition	without

Compound: A compound is a word composed of more than one free morpheme.

Terminology



Precision (positive predictive value) and recall (sensitivity)

Precision is the fraction of retrieved documents that are relevant to the query:

$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

Recall is the fraction of the relevant documents that are successfully retrieved:

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Terminology



You implement a software application for retrieving documents containing one specific word (“cat” in the examples) and its inflectional forms in a corpus.

There are 24 documents in the corpus of 1000 documents, containing “cat” (**24 relevant documents**).

Example 1: Your application retrieves 36 documents (**36 retrieved documents**), out of which 16 contain the word “cat” (**16 relevant documents** ∩ **retrieved documents**), and 20 do not contain it.

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

Precision = 16 / 36

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Recall = 16 / 24

Terminology



You implement a software application for retrieving documents containing one specific word (“cat” in the examples) and its inflectional forms in a corpus.

There are 24 documents in the corpus of 1000 documents, containing “cat” (**24 relevant documents**).

Example 2: Your application retrieves all the 1000 documents (**1000 retrieved documents**), out of which 24 contain the word “cat” (**24 relevant documents** ∩ **retrieved documents**), and 1986 do not contain it.

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

Precision = 24 / 1000

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Recall = 24 / 24

Terminology



You implement a software application for retrieving documents containing one specific word (“cat” in the examples) and its inflectional forms in a corpus.

There are 24 documents in the corpus of 1000 documents, containing “cat” (**24 relevant documents**).

Example 3: Your application retrieves 2 documents (**2 retrieved documents**), out of which 2 contain the word “cat” (**2 relevant documents** ∩ **retrieved documents**), and 0 do not contain it.

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

Precision = 2 / 2

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Recall = 2 / 24

Terminology



You implement a software application for retrieving documents containing one specific word (“cat” in the examples) and its inflectional forms in a corpus.

There are 24 documents in the corpus of 1000 documents, containing “cat” (**24 relevant documents**).

Example 4: Your application retrieves 24 documents (**24 retrieved documents**), out of which 24 contain the word “cat” (**24 relevant documents** ∩ **retrieved documents**), and 0 do not contain it.

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

Precision = 24 / 24

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Recall = 24 / 24